

ERPsim for SAP HANA

OData Simulation Service - Reference Guide

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GET STARTED: ACQUIRING DATA

With the ERPSim 2023 Release, OData is the only method for accessing data.

Solution / Tool	OData	Supported OS
SAP Analytics Cloud	Yes	Windows
SAP Lumira Discovery 2.3	Yes	Windows
SAP Lumira (Previous versions)	Yes	Windows
SAP Predictive Analytics (3.3.9)	Yes	Windows
Microsoft Excel (except Excel 2016)	Yes	Windows, macOS
Microsoft PowerBI	Yes	Windows
Tableau Desktop	Yes	Windows, macOS
Tableau Public (2020.3.0 and higher)	No	Windows, macOS
Tableau Public (Previous versions)	Yes	Windows, macOS
Google Sheets	No	Any

To review the process of connecting to each of the above solutions, please review their individual guide from the *Tool-Specific Instructions* section on the *Analytics with ERPSim for SAP S/4HANA* page of our website: <https://erpsim.hec.ca/en/node/419>.

ODATA VIEWS AND VIRTUAL TIME TAGGING

The data used to build the OData views comes from a snapshot of documents and tables that are tagged at the end of each step. Some data, such as the inventory or the current pricing conditions, are therefore labeled for the next step.

Current Step Example: At the end of Step 6, the financial postings processed during this step are going to be tagged as “Step 6”.

Next Step Example: At the end of Step 6, your inventory is going to be tagged as the starting inventory of Step 7, even though Step 7 has not been played yet.

The table on the right shows the tagging rule and the games supported for each of the views.

Please note that the “Carbon Emissions” view is only available for the Manufacturing and Logistics Sustainability scenarios.

	Current Step	Next Step	Manufacturing	Logistics	Maple
OData Views					
BOM_Changes	✓		✓		
Carbon_Emissions	✓		✓	✓	
Company_Valuation	✓		✓	✓	✓
Current_Game_Rules	✓		✓	✓	✓
Current_Inventory	✓		✓	✓	✓
Current_Inventory_KPI	✓		✓	✓	✓
Current_Pricing_Conditions		✓	✓	✓	✓
Current_Suppliers_Prices	✓		✓	✓	✓
Financial_Postings	✓		✓	✓	✓
Independent_Requirements	✓		✓	✓	✓
Inventory		✓	✓	✓	✓
Market	✓		✓	✓	✓
Marketing_Expenses		✓	✓		✓
Production	✓		✓		
Production_Orders	✓		✓		
Purchase_Orders	✓		✓	✓	✓
Sales	✓		✓	✓	✓
Stock_Transfers	✓		✓	✓	

ODATA VIEWS DETAILS

BOM_Changes

Description	Tracks BOM changes and their new recipes.
Primary Use	Combine this view with “Sales” and/or “Purchase_Orders” to analyze how BOM changes affect your costs and the market demand.
Relevant Transaction(s) in SAP	Validated BOM Change (ZCS02)
Scope (ERPsim games)	Manufacturing (Extended and Advanced)

Measures

Label	Technical Name	Aggregation	Description
Quantity	QUANTITY	Sum	Quantity of components required to produce one unit of finished goods.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow ‘order by’ clauses.
Plant	PLANT	Text	Defines the plant for which the data will be provided.
Simulation Round	SIM_ROUND	Integer	Round component of the virtual date in ERPsim.
Simulation Step	SIM_STEP	Integer	Step component of the virtual date in ERPsim.
Simulation Date	SIM_DATE	Text	Concatenation of Simulation Round and Simulation Step.

Label	Technical Name	Type	Description
Simulation Period	SIM_PERIOD	Integer	A period is a cluster of 5 steps.
Elapsed Steps	SIM_ELAPSED_STEPS	Integer	Number of steps since the beginning of the simulation.
Material Number	MATERIAL_NUMBER	Text	Unique number that identifies a product.
Component	COMPONENT	Text	Material Number of the component (raw material or packaging).
Material Description	MATERIAL_DESCRIPTION	Text	Complete description of the product, including its size (if applicable).
Unit	UNIT	Text	Refers to the unit of measure defined in the master data for the product.

Carbon_Emissions

Description	Tracks the company's activities generating CO2e emissions.
Primary Use	Analyzes your carbon-generating activities to identify potential issues and opportunities for improvement.
Relevant Transaction(s) in SAP	Carbon Footprint Analytic Tile (SSB Tile)
Scope (ERPsim games)	Logistics (Sustainability), Manufacturing (Sustainability)

Measures

Label	Technical Name	Aggregation	Description
Total CO2 Emissions	TOTAL_CO2_EMISSIONS	Sum	Total CO2e emissions generated by this activity.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Company Code	COMPANY_CODE	Text	Defines the company code for which the data will be provided.
Simulation Round	SIM_ROUND	Integer	Round component of the virtual date in ERPsim.
Simulation Step	SIM_STEP	Integer	Step component of the virtual date in ERPsim.
Simulation Date	SIM_DATE	Text	Concatenation of Simulation Round and Simulation Step.
Simulation Period	SIM_PERIOD	Integer	A period is a cluster of 5 steps.

Label	Technical Name	Type	Description
Elapsed Steps	SIM_ELAPSED_STEPS	Integer	Number of steps since the beginning of the simulation.
Type	TYPE	Text	Type of activity generating carbon emissions.
Subtype	SUBTYPE	Text	Gives further information on the type of activity. The following types/subtypes can be listed: • Goods Movement ○ Goods Receipt ○ Internal Transfers ○ Deliveries • Overhead ○ Purchased Energy ○ Other • Overstock ○ CO2e from additional storage • Products Purchased ○ Upstream CO2e per supplier • Production ○ Units Produced ○ Setup Time • Investments ○ Capacity Improvement ○ Lean Manufacturing Program ○ Sustainable Manufacturing ○ Freight Fleet Improvements
Origin	ORIGIN	Text	Specifies the origin when the Type is “Goods Movement” or “Products Purchased”. The origin can either be a supplier code or a storage location.

Label	Technical Name	Type	Description
Destination	DESTINATION	Text	Specifies the destination when the Type is “Goods Movement” or “Products Purchased”. The destination will always be a storage location.
Product	PRODUCT	Text	Specifies the product generating carbon emissions when the Type is “Products Purchased” or “Production”. If the Type is “Overstock”, this field will specify which kind of storage was overstocked (e.g., Packaging).
Document	DOCUMENT	Integer	Material Document linked to the activity. May be empty if no document is linked to the activity.
Scope	SCOPE	Integer	Classification of the carbon emissions, according to GHG Protocol Corporate Standard¹. In the ERPsim simulation, the scopes are defined as the following: • Scope 1 - Direct emissions from owned or controlled sources: ○ Internal goods movements ○ Deliveries to customers ○ Overstocking carbon emissions ○ Setup time (Manufacturing only) ○ Units produced (Manufacturing only) ○ Pickup at suppliers’ sites (Manufacturing only) • Scope 2 - Indirect emissions from the generation of purchased energy: ○ Purchased energy (as overheads) • Scope 3 - Other Indirect emissions (upstream and downstream): ○ Goods receipts from a supplier ○ Products purchased from a supplier ○ Other overheads ○ Investments (Manufacturing only) ○ Customer pickup (Manufacturing only)

¹ https://ghgprotocol.org/sites/default/files/standards_supporting/FAQ.pdf

Label	Technical Name	Type	Description
CO2e Emissions	CO2E_EMISSIONS	Float	CO2e emissions generated by one instance of an activity. The Emissions column may differ from the Total CO2 Emissions column when the activity is repeated multiple times in the same virtual day (e.g., multiple units produced, multiple deliveries).

Company_Valuation

Description	Tracks the company valuation and all accounts used in its calculation.
Primary Use	Identify periods of sharp valuation changes and compare with other views to understand their causes.
Relevant Transaction(s) in SAP	Company Valuation Report (ZVAL), Company Valuation Analytic Tile (SSB Tile)
Scope (ERPsim games)	All

Measures

Label	Technical Name	Aggregation	Description
Bank Cash Account	BANK_CASH_ACCOUNT	Sum	Amount in the account “113300 Bank Cash Account”.
Accounts Receivable	ACCOUNTS_RECEIVABLE	Sum	Amount in the account “140000 Customer - Domestic Receivables”.
Bank Loan	BANK_LOAN	Sum	Amount in the account “113101 Bank Loan”.
Accounts Payable	ACCOUNTS_PAYABLE	Sum	Amount in the account “160000 Accounts payable-domestic”.
Profit	PROFIT	Sum	Total profit. The sum of all Income Statement accounts.
Setup Time Investment	SETUP_TIME_INVESTMENT	Sum	Amount in the account “478000 Lean Manufacturing Program Expense”.
Debt Loading	DEBT_LOADING	Sum	Company’s debt load (i.e., Bank Cash Account + Accounts Receivable - Bank Loan - Accounts Payable).
Company Risk Rate (%)	COMPANY_RISK_RATE_PCT	Avg	Risk of losses from changes in the interest rate. The Company Risk Rate is based on the company’s Credit Rating.
Market Risk Rate (%)	MARKET_RISK_RATE_PCT	Avg	The Market Risk Rate is based on historical stock market average returns. In the simulation, the Market Risk Rate is always 7%.

Label	Technical Name	Aggregation	Description
Company Valuation	COMPANY_VALUATION	Sum	Metric to allow for consistent comparison of team performance based on profit & risk. For the complete formula, please refer to the note on the following page.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Company Code	COMPANY_CODE	Text	Defines the company code for which the data will be provided.
Simulation Round	SIM_ROUND	Integer	Round component of the virtual date in ERPsim.
Simulation Step	SIM_STEP	Integer	Step component of the virtual date in ERPsim.
Simulation Date	SIM_DATE	Text	Concatenation of Simulation Round and Simulation Step.
Simulation Period	SIM_PERIOD	Integer	A period is a cluster of 5 steps.
Elapsed Steps	SIM_ELAPSED_STEPS	Integer	Number of steps since the beginning of the simulation.
Credit Rating	CREDIT_RATING	Text	Evaluation of the credit risk of the company. The Credit Rating is based on the company's Debt Loading.
Currency	CURRENCY	Text	Key identifying the currency used for monetary values.

Formula for Company Valuation

$$\left(\frac{\left(\left(\frac{(\text{Profit} + \text{Setup Time Investment})}{\text{Elapsed Steps}} \right) * (x * y) \right) - \text{Setup Time Investment}}{\text{Company Risk Rate} + \text{Market Risk Rate}} \right) / 100$$

Where:

- x = Maximum number of rounds for the current game scenario
- y = Maximum number of steps per round for the current game scenario
- Setup Time Investment is a positive value
- Setup Time Investment is only possible in the Manufacturing version of the game. You can replace this value by 0 for all other games.

Current_Game_Rules

Description	Lists all relevant game rules and organizational elements for the given game and scenario.
Primary Use	Use this view to clarify specific rules, like how you would use the Job Aid.
Relevant Transaction(s) in SAP	Most transactions
Scope (ERPsim games)	All

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Company Code	COMPANY_CODE	Text	Defines the company code for which the data will be provided.
Game	GAME	Text	Defines the game for which the rules apply.
Scenario	SCENARIO	Text	Defines the scenario for which the rules apply.
Category	CATEGORY	Text	Gives information on the scope of the rule provided. The following categories can be listed: • Organizational_Element: The organizational elements used during the game. • Product: All products available in the game • Rules: Other general rules of the game
Element	ELEMENT	Text	Defines the element impacted by the rule provided.
Detail	DETAIL	Text	Gives further information on the element impacted by the rule provided.
Value	VALUE	Text	Value of the rule provided, or name of the element provided.

Current_Inventory

Description	Tracks the current inventory for the materials relevant to the game scenario.
Primary Use	Use in dashboards with alerts or indicators to notify users when inventory is too high or too low.
Relevant Transaction(s) in SAP	Inventory Report (ZMB52)
Scope (ERPsims games)	All

Measures

Label	Technical Name	Aggregation	Description
Stock	STOCK	Sum	Current product quantity in stock.
Restricted	RESTRICTED	Sum	Current product quantity reserved for consumption in production orders.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Plant	PLANT	Text	Defines the plant for which the data will be provided.
Material Number	MATERIAL_NUMBER	Text	Unique number that identifies a product.
Material Description	MATERIAL_DESCRIPTION	Text	Complete description of the product, including its size (if applicable).
Storage Location	STORAGE_LOCATION	Text	Indicates the storage location in which the product is stored.
Unit	UNIT	Text	Refers to the unit of measure defined in the master data for the product.

Current_Inventory_KPI

Description	Provides key indicators to better manage inventory, stock transfer and forecast.
Primary Use	Use in dashboards with alerts or indicators to notify users to reorder new products.
Relevant Transaction(s) in SAP	Inventory Report (ZMB52), Sales Order Reports (ZVA05 and ZVC2)
Scope (ERPsim games)	All

Measures

Label	Technical Name	Aggregation	Description
Current Inventory	CURRENT_INVENTORY	Sum	Current product quantity in stock.
Quantity Sold	QUANTITY_SOLD	Sum	Represents the total product quantity sold to customers.
Nb Steps Available	NB_STEPS_AVAILABLE	Sum	Number of steps for which the product was in inventory. Can be used with Elapsed Steps to analyse your stockout frequency.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Plant	PLANT	Text	Defines the plant for which the data will be provided.
Storage Location	STORAGE_LOCATION	Text	Indicates the storage location in which the product is stored.
Material Number	MATERIAL_NUMBER	Text	Unique number that identifies a product.
Material Description	MATERIAL_DESCRIPTION	Text	Complete description of the product, including its size (if applicable).

Label	Technical Name	Type	Description
Unit	UNIT	Text	Refers to the unit of measure defined in the master data for the product.
Elapsed Steps	SIM_ELAPSED_STEPS	Integer	Number of steps since the beginning of the simulation.

Current_Pricing_Conditions

Description	Provides the current selling prices of your products.
Primary Use	Compare pricing with the "Sales" views to evaluate its impact on market demand.
Relevant Transaction(s) in SAP	Change Price List (VK32)
Scope (ERPsim games)	All

Measures

Label	Technical Name	Aggregation	Description
Price	PRICE	Sum	Selling price of the product for the given distribution channel and virtual day.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Sales Organization	SALES_ORGANIZATION	Text	Defines the sales organization for which the data will be provided.
Material Number	MATERIAL_NUMBER	Text	Unique number that identifies a product.
Material Description	MATERIAL_DESCRIPTION	Text	Complete description of the product, including its size (if applicable).
Distribution Channel	DISTRIBUTION_CHANNEL	Text	Unique distribution channel identifier.
DC Name	DC_NAME	Text	Name of the distribution channel.
Currency	CURRENCY	Text	Key identifying the currency used for monetary values.

Current_Suppliers_Prices

Description	Provides the current prices of purchasable goods.
Primary Use	Use in dashboards with alerts or indicators to flag when a supplier's product is unusually cheap or expensive.
Relevant Transaction(s) in SAP	Procurement Sourcing (ZME13)
Scope (ERPsims games)	All

Measures

Label	Technical Name	Aggregation	Description
Net Price	NET_PRICE	Max	Price at which one unit of the material is available.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Purchasing Organization	PURCHASING_ORGANIZATION	Text	Defines the purchasing organization for which the data will be provided.
Material Number	MATERIAL_NUMBER	Text	Unique number that identifies a product.
Material Description	MATERIAL_DESCRIPTION	Text	Complete description of the product, including its size (if applicable).
Vendor Code	VENDOR_CODE	Text	Unique vendor identifier.
Vendor Name	VENDOR_NAME	Text	Name of the vendor.
Currency	CURRENCY	Text	Key identifying the currency used for monetary values.

Financial_Postings

Description	Provides live financial statements and day-to-day accounting transactions. Allows G/L account analysis using time series. Amounts are provided as real (debit+/credit- or debit-/credit+) and absolute values (requires the use of the debit/credit indicator dimension).
Primary Use	Track all company investments and non-operational expenses.
Relevant Transaction(s) in SAP	Financial Statements (F.01)
Scope (ERPsim games)	All

Measures

Label	Technical Name	Aggregation	Description
Amount	AMOUNT	Sum	Posting amount, where debit postings are positive and credit postings are negative.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Company Code	COMPANY_CODE	Text	Defines the company code for which the data will be provided.
Simulation Round	SIMULATION_ROUND	Integer	Round component of the virtual date in ERPsim.
Simulation Step	SIMULATION_STEP	Integer	Step component of the virtual date in ERPsim.
Simulation Date	SIM_DATE	Text	Concatenation of Simulation Round and Simulation Step.

Label	Technical Name	Type	Description
Simulation Period	SIM_PERIOD	Integer	A period is a cluster of 5 steps.
Elapsed Steps	SIM_ELAPSED_STEPS	Integer	Number of steps since the beginning of the simulation.
Account Number	GL_ACCOUNT_NUMBER	Text	Unique General Ledger account identifier.
Account Name	GL_ACCOUNT_NAME	Text	Label describing the General Ledger account.
Financial Statements (1) <i>FS Level 1</i> (2) <i>FS Level 2</i> (3) <i>FS Level 3</i> (4) <i>FS Level 4</i>	Financial_Statements (1) <i>FS_LEVEL_1</i> (2) <i>FS_LEVEL_2</i> (3) <i>FS_LEVEL_3</i> (4) <i>FS_LEVEL_4</i>	Text	Allows the drill-down of values through the financial statements. Starts with the highest level (FS Level 1), with up to 4 levels.
Debit/Credit Indicator	DEBIT_CREDIT_INDICATOR	Text	Indicates if the posting to the G/L account was made as a debit or a credit.
Currency	CURRENCY	Text	Key identifying the currency used to maintain the account balance.

Independent_Requirements

Description	Tracks changes in independent requirements.
Primary Use	Track all independent requirements plans & compare with other views (e.g., Sales) to determine the optimal procurement strategy.
Relevant Transaction(s) in SAP	Create Planned Independent Requirements (MD61)
Scope (ERPsim games)	All

Measures

Label	Technical Name	Aggregation	Description
Quantity	QUANTITY	Sum	Represents the newly forecasted product quantities.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Plant	PLANT	Text	Defines the plant for which the data will be provided.
Simulation Round	SIM_ROUND	Integer	Round component of the virtual date in ERPsim.
Simulation Step	SIM_STEP	Integer	Step component of the virtual date in ERPsim.
Simulation Date	SIM_DATE	Text	Concatenation of Simulation Round and Simulation Step.
Simulation Period	SIM_PERIOD	Integer	A period is a cluster of 5 steps.
Elapsed Steps	SIM_ELAPSED_STEPS	Integer	Number of steps since the beginning of the simulation.

Label	Technical Name	Type	Description
Material Number	MATERIAL_NUMBER	Text	Unique number that identifies a product.
Unit	UNIT	Text	Refers to the unit of measure defined in the master data for the product.

Inventory

Description	Tracks opening inventory balance over time for each material and storage location.
Primary Use	Maintain a history of storage utilization to monitor capacity usage and identify overstock periods.
Relevant Transaction(s) in SAP	Inventory Report (ZMB52)
Scope (ERPsim games)	All

Measures

Label	Technical Name	Aggregation	Description
Inventory Opening Balance	INVENTORY_OPENING_BALANCE	Sum	Stock quantity at the beginning of each day.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Plant	PLANT	Text	Defines the plant for which the data will be provided.
Simulation Round	SIM_ROUND	Integer	Round component of the virtual date in ERPsim.
Simulation Step	SIM_STEP	Integer	Step component of the virtual date in ERPsim.
Simulation Date	SIM_DATE	Text	Concatenation of Simulation Round and Simulation Step.
Simulation Period	SIM_PERIOD	Integer	A period is a cluster of 5 steps.
Elapsed Steps	SIM_ELAPSED_STEPS	Integer	Number of steps since the beginning of the simulation.

Label	Technical Name	Type	Description
Storage Location	STORAGE_LOCATION	Text	Indicates the storage location in which the product is stored.
Material Number	MATERIAL_NUMBER	Text	Unique number that identifies a product.
Material Description	MATERIAL_DESCRIPTION	Text	Complete description of the product, including its size (if applicable).
Unit	UNIT	Text	Refers to the unit of measure defined in the master data for the product.

Market

Description	Analyzes market shares based on sales quantity or net value.
Primary Use	Compare your company's sales and prices with the whole market.
Relevant Transaction(s) in SAP	Price Market Report (ZMARKET)
Scope (ERPsim games)	All

Measures

Label	Technical Name	Aggregation	Description
Quantity	QUANTITY	Sum	Total product quantity sold to customers for the period.
Average Price	AVERAGE_PRICE	Avg	Average selling price of the goods sold within a combination of a distribution channel / region.
Net Value	NET_VALUE	Sum	Value of the sales orders that were generated. The net value equals the quantity sold of each item multiplied by its respective unit price.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Company Code	COMPANY_CODE	Text	Defines the company code for which the data will be provided.
Sales Organization	SALES_ORGANIZATION	Text	Defines the unit to which the data are attached to. Only one unit is currently available:

Label	Technical Name	Type	Description
			“Market” returns data for the whole market, including your own company. Subtract the data from the “Sales” view to isolate the data from your competitors.
Simulation Round	SIM_ROUND	Integer	Round component of the virtual date in ERPsim.
Simulation Period	SIM_PERIOD	Integer	A period is a cluster of 5 steps. Market reports are released at the end of each period.
Material Description	MATERIAL_DESCRIPTION	Text	Complete description of the product, including its size (if applicable).
Distribution Channel	DISTRIBUTION_CHANNEL	Text	Unique distribution channel identifier.
Area	AREA	Text	Defines the region in which the sales were made (North, South, West).
Unit	UNIT	Text	Refers to the unit of measure defined in the master data for the product.
Currency	CURRENCY	Text	Key identifying the currency used for monetary values.

Marketing_Expenses

Description	Tracks the daily investments in marketing per product and region. In Round 1 Step 1, all products are tagged to ensure a complete visualization; in every other step, only products with marketing expenses are tagged.
Primary Use	Compare with the "Sales" view to assess marketing's impact on sales.
Relevant Transaction(s) in SAP	Marketing Expense Planning (ZADS)
Scope (ERPsim games)	Manufacturing (All), Maple

Measures

Label	Technical Name	Aggregation	Description
Amount	AMOUNT	Sum	Daily value of the marketing investment.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Sales Organization	SALES_ORGANIZATION	Text	Defines the sales organization for which the data will be provided.
Simulation Round	SIM_ROUND	Integer	Round component of the virtual date in ERPsim.
Simulation Step	SIM_STEP	Integer	Step component of the virtual date in ERPsim.
Simulation Date	SIM_DATE	Text	Concatenation of Simulation Round and Simulation Step.
Simulation Period	SIM_PERIOD	Integer	A period is a cluster of 5 steps.

Label	Technical Name	Type	Description
Elapsed Steps	SIM_ELAPSED_STEPS	Integer	Number of steps since the beginning of the simulation.
Material Number	MATERIAL_NUMBER	Text	Unique number that identifies a product.
Material Description	MATERIAL_DESCRIPTION	Text	Complete description of the product, including its size (if applicable).
Area	AREA	Text	Defines the region in which the marketing investments were made (North, South, West).
Currency	CURRENCY	Text	Key identifying the currency used for monetary values.

Production

Description	Tracks production yield per product daily. In Round 1 Step 1, all products are tagged to ensure a complete visualization; in every other step, only products with production are tagged.
Primary Use	Monitor daily productivity and track the exact production dates of your products.
Relevant Transaction(s) in SAP	Production Report (ZCOOIS)
Scope (ERPsim games)	Manufacturing (All)

Measures

Label	Technical Name	Aggregation	Description
Yield	YIELD	Sum	Quantity of finished goods produced.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Company Code	COMPANY_CODE	Text	Defines the company code for which the data will be provided.
Simulation Round	SIM_ROUND	Integer	Round component of the virtual date in ERPsim.
Simulation Step	SIM_STEP	Integer	Step component of the virtual date in ERPsim.
Simulation Date	SIM_DATE	Text	Concatenation of Simulation Round and Simulation Step.
Simulation Period	SIM_PERIOD	Integer	A period is a cluster of 5 steps.

Label	Technical Name	Type	Description
Elapsed Steps	SIM_ELAPSED_STEPS	Integer	Number of steps since the beginning of the simulation.
Material Number	MATERIAL_NUMBER	Text	Unique number that identifies a product.
Material Description	MATERIAL_DESCRIPTION	Text	Complete description of the product, including its size (if applicable).
Unit	UNIT	Text	Refers to the unit of measure defined in the master data for the product.

Production_Orders

Description	Tracks all production orders documents generated and their status.
Primary Use	Evaluate your production schedule to optimize order releases and minimize setup time.
Relevant Transaction(s) in SAP	Production Report (ZCOOIS)
Scope (ERPsim games)	Manufacturing (All)

Measures

Label	Technical Name	Aggregation	Description
Target Quantity	TARGET_QUANTITY	Sum	The total quantity to be produced to complete the production order.
Confirmed Quantity	CONFIRMED_QUANTITY	Sum	Current quantity produced for the production order.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Company Code	COMPANY_CODE	Text	Defines the company code for which the data will be provided.
Simulation Round	SIM_ROUND	Integer	Round component of the virtual date in ERPsim.
Simulation Step	SIM_STEP	Integer	Step component of the virtual date in ERPsim.
Simulation Date	SIM_DATE	Text	Concatenation of Simulation Round and Simulation Step.
Simulation Period	SIM_PERIOD	Integer	A period is a cluster of 5 steps.

Label	Technical Name	Type	Description
Elapsed Steps	SIM_ELAPSED_STEPS	Integer	Number of steps since the beginning of the simulation.
Production Order	PRODUCTION_ORDER	Text	Production order document number.
Material Number	MATERIAL_NUMBER	Text	Unique number that identifies a product.
Material Description	MATERIAL_DESCRIPTION	Text	Complete description of the product, including its size (if applicable).
Begin Round	BEGIN_ROUND	Integer	Round on which the production will start.
Begin Step	BEGIN_STEP	Integer	Step on which the production will start.
End Round	END_ROUND	Integer	Round on which the production will end.
End Step	END_STEP	Integer	Step on which the production will end.
Unit	UNIT	Text	Refers to the unit of measure defined in the master data for the product.
Setup Time	SETUP_TIME	Integer	Amount of time required to set up the production line (in hours). The higher the Setup Time is, the less daily production you will be able to carry out.

Purchase_Orders

Description	Tracks all purchase orders documents generated and their status.
Primary Use	Monitor incoming purchase orders and their sources of supply.
Relevant Transaction(s) in SAP	Purchase Order Tracking (ZME2N)
Scope (ERPsim games)	All

Measures

Label	Technical Name	Aggregation	Description
Quantity	QUANTITY	Sum	Product quantity in the purchase order.
Unit Price	UNIT_PRICE	Sum	Unit price of the product for that purchase order.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Company Code	COMPANY_CODE	Text	Defines the company code for which the data will be provided.
Simulation Round	SIM_ROUND	Integer	Round component of the virtual date in ERPsim.
Simulation Step	SIM_STEP	Integer	Step component of the virtual date in ERPsim.
Simulation Date	SIM_DATE	Text	Concatenation of Simulation Round and Simulation Step.
Simulation Period	SIM_PERIOD	Integer	A period is a cluster of 5 steps.

Label	Technical Name	Type	Description
Elapsed Steps	SIM_ELAPSED_STEPS	Integer	Number of steps since the beginning of the simulation.
Purchasing Order	PURCHASING_ORDER	Text	Purchase order document number.
Vendor	VENDOR	Text	Unique vendor identifier.
Material Number	MATERIAL_NUMBER	Text	Unique number that identifies a product.
Material Description	MATERIAL_DESCRIPTION	Text	Complete description of the product, including its size (if applicable).
Status	STATUS	Text	Status of the delivery.
Unit	UNIT	Text	Refers to the unit of measure defined in the master data for the product.
Goods Receipt Round	GOODS_RECEIPT_ROUND	Integer	Round on which the purchase order will be received.
Goods Receipt Step	GOODS_RECEIPT_STEP	Integer	Step on which the purchase order will be received.
Goods Receipt Date	GOODS_RECEIPT_DATE	Text	Concatenation of Goods Receipt Round and Goods Receipt Step.

Sales

Description	Tracks all sales documents and allows you to explore and gain insight from your sales.
Primary Use	Analyze sales data to identify market trends and highlight top- and low-performing products.
Relevant Transaction(s) in SAP	Sales Order Reports (ZVA05 and ZVC2)
Scope (ERPsim games)	All

Measures

Label	Technical Name	Aggregation	Description
Quantity	QUANTITY	Sum	Represents the sales order item quantity that was purchased by the customer.
Quantity Delivered	QUANTITY_DELIVERED	Sum	Represents the sales order item quantity that was delivered to the customer.
Net Value	NET_VALUE	Sum	Value of the sales orders that were generated. The net value equals the quantity sold of each item multiplied by its respective unit price.
Cost	COST	Sum	Represents the cost of the product based on its valuation.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Sales Organization	SALES_ORGANIZATION	Text	Defines the sales organization for which the data will be provided.
Simulation Round	SIM_ROUND	Integer	Round component of the virtual date in ERPsim.

Label	Technical Name	Type	Description
Simulation Step	SIM_STEP	Integer	Step component of the virtual date in ERPsim.
Simulation Date	SIM_DATE	Text	Concatenation of Simulation Round and Simulation Step.
Simulation Period	SIM_PERIOD	Integer	A period is a cluster of 5 steps.
Elapsed Steps	SIM_ELAPSED_STEPS	Integer	Number of steps since the beginning of the simulation.
Sales Order Number	SALES_ORDER_NUMBER	Text	Sales order document number.
Line Item	LINE_ITEM	Text	Line item in the sales order document.
Storage Location	STORAGE_LOCATION	Text	Indicates the storage location in which the product is stored.
Area	AREA	Text	Defines the region in which the sales were made (North, South, West).
Distribution Channel	DISTRIBUTION_CHANNEL	Text	Unique distribution channel identifier.
Material Number	MATERIAL_NUMBER	Text	Unique number that identifies a product.
Material Description	MATERIAL_DESCRIPTION	Text	Complete description of the product, including its size (if applicable).
Unit	UNIT	Text	Refers to the unit of measure defined in the master data for the product.
Net Price	NET PRICE	Decimal	Average item price per item that was sold to customers.
Currency	CURRENCY	Text	Key identifying the currency used for monetary values.

Stock_Transfers

Description	Tracks the stock transfers between your main warehouse and your regional warehouses.
Primary Use	Compare with the "Sales" and "Inventory" views to identify stockouts and optimize product availability.
Relevant Transaction(s) in SAP	Stock Transfer (ZMB1B)
Scope (ERPsim games)	Logistics (All), Manufacturing (Sustainability and Advanced)

Measures

Label	Technical Name	Aggregation	Description
Quantity	QUANTITY	Sum	Product quantity transferred to the regional warehouse.

Dimensions

Label	Technical Name	Type	Description
Row ID	ROW_ID	Integer	Unique row ID generated to allow 'order by' clauses.
Plant	PLANT	Text	Defines the plant for which the data will be provided.
Simulation Round	SIM_ROUND	Integer	Round component of the virtual date in ERPsim.
Simulation Step	SIM_STEP	Integer	Step component of the virtual date in ERPsim.
Simulation Date	SIM_DATE	Text	Concatenation of Simulation Round and Simulation Step.
Simulation Period	SIM_PERIOD	Integer	A period is a cluster of 5 steps.
Elapsed Steps	SIM_ELAPSED_STEPS	Integer	Number of steps since the beginning of the simulation.

Label	Technical Name	Type	Description
Storage Location	STORAGE_LOCATION	Text	Indicates the target storage location on which the goods movement applies.
Material Number	MATERIAL_NUMBER	Text	Unique number that identifies a product.
Material Description	MATERIAL_DESCRIPTION	Text	Complete description of the product, including its size (if applicable).
Unit	UNIT	Text	Refers to the unit of measure defined in the master data for the product.